

Pranshu Kumar Singh

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I am an experienced software engineer with 8 years of progressive expertise spanning application ownership, change management, application support, and system administration. For the past 7 years, I have specialized in DevOps, leveraging Azure cloud and modern DevOps tools including Docker, Kubernetes, Terraform, Prometheus, and Grafana. My proficiency extends to Linux operating systems and incident management using ServiceNow. I bring a holistic, hands-on approach to infrastructure automation, deployment, monitoring, and continuous improvement in fast-paced enterprise environments.

Experience

August 2025 – Present
Lead Engineer – Feuji Systems

Designation – Senior Software Engineer Client– NorthenTrust Bank

As a Senior Software Engineer – DevOps at Fuji System, I managed end-to-end Azure cloud infrastructure deployments utilizing Terraform, Azure DevOps, Azure Boards, and GitHub. My responsibilities included provisioning and maintaining cloud resources as per evolving business requirements, implementing CI/CD pipelines, configuring robust monitoring and alerting solutions, and ensuring seamless application testing across development, UAT, staging, and production environments. I actively participated in sprint-based project execution by addressing user stories, automating workflows, troubleshooting deployments, and collaborating with cross-functional teams to deliver high-quality, secure, and scalable solutions on the Azure cloud platform.

April 2023 – Aug 2025

Senior Specialist - HCL Tech

Designation - Senior Devops Consultant

Client - State Farm,USA

As a DevOps Engineer specialised in BFSI (Banking, Financial Services, and Insurance) clients at HCL Technologies, at the forefront of transforming traditional financial systems into agile, automated, and secure environments. collaborating closely with both development and operations teams to streamline processes, enhance reliability, and ensure compliance with industry standards and regulations.

Key Responsibilities:

- **DevOps Strategy Development:** Developing and implementing DevOps strategies tailored to the unique requirements and challenges of BFSI clients, ensuring alignment with business objectives and regulatory standards.
- **Continuous Integration/Continuous Deployment (CI/CD):** Designing and maintaining robust CI/CD pipelines to automate the build, test, and deployment processes, enabling rapid and reliable delivery of software updates and enhancements.
- **Infrastructure as Code (IaC):** Utilising tools such as Terraform, Ansible, and Puppet to provision and manage infrastructure resources programmatically, ensuring scalability, consistency, and efficiency across BFSI environments.
- **Security and Compliance:** Implementing security best practices and compliance measures (e.g., PCI DSS, GDPR, SOX) within the DevOps pipeline and infrastructure to safeguard sensitive financial data and mitigate risks.
- **Monitoring and Incident Response:** Establishing comprehensive monitoring solutions to proactively detect and address performance issues, security threats, and

compliance violations, ensuring high availability and reliability of BFSI systems using Prometheus, Grafana, Splunk and Dynatrace

- **Automation and Orchestration:** Identifying opportunities for automation and orchestration across the software delivery lifecycle, leveraging tools like Jenkins, GitLab, and Bamboo to streamline processes and reduce manual intervention.

- **Containerisation and Orchestration:** Deploying and managing containerised applications using platforms such as Docker and Kubernetes, optimising resource utilisation and facilitating seamless scalability in BFSI environments.

- **Collaboration and Communication:** Foster effective collaboration and communication among cross-functional teams, including developers, QA engineers, system administrators, and business stakeholders, to facilitate the adoption of DevOps practices within BFSI projects.

- **Technical Support and Troubleshooting:** Providing technical support and troubleshooting assistance to resolve issues and ensure the smooth operation of BFSI systems, leveraging your expertise in cloud platforms, networking, and distributed systems.

- **Knowledge Sharing and Mentoring:** Stay abreast of emerging technologies, industry trends, and best practices in DevOps and BFSI domains, and share your knowledge and insights with colleagues through mentorship, training sessions, and knowledge-sharing forums.

December 2021 – April 2023

LEAD ASSOCIATE, GENPACT

Designation - DevOps Consultant

Client- Airbnb

- Working as DevOps Engineer in Genpact Being Airbnb as client playing a number of key responsibilities integrating the project functions and resources across the product life cycle, right from planning, building, testing, and deployment.

Key Responsibilities:

- Configure & Manage Applications on AWS platform
- Setup DevOps infrastructure for Applications
- Carry out Deployments & Releases using GitHub, Jenkins(CI/CD), Docker & Chef for all environments
- Maintenance & Upgradation of Dev, Test, Pre-prod, Stage & Prod Windows & Linux Application servers
- Manage Application Pods using Kubernetes
- Supervise Infrastructure Management for hosted application and services
- Render L2.5 Application Support for UPC & Camunda
- Execute OS Patching for Windows & Linux servers using BigFix
- Update SSL Certificates & Licenses for all servers
- Implement changes & upgrades in Database using PostgreSQL & SQL Server Management Studio
- Perform Audits on other Applications post Production Release
- Ticket Handling using Jira & ServiceNow
- Track & Resolve Alerts from Dynatrace, Prometheus, Grafana, PagerDuty, Datadog, logicmonitor

April 2018 – December 2021

Software Engineer, Tech Mahindra

Designation- Associate Software Engineer **Client-** AT&T (USA)

Software Engineer with 3 years of experience working with Tech Mahindra for USA Telecom Giant AT&T being client under role of Production Operations. With ample experience of handling Production issue in real time and maintaining perfect balance among as stake holders for the project.

- AT&T (USA) Application Production Support.
- Committed to 99.99% uptime SLA Compliance

- Successfully deployed major/minor Projects in Production Environment
- Coordinated with SA, DBA, Application Developer, and other stake holders for timely resolution of issues with minimal or no downtime
- Supported periodic UNIX/LINUX and database server patching

Education

June 2017

B. TECH (MAE), Amity university, Lucknow

- Always being meritorious student threw out the academic period.
 - Published research paper for major and minor projects.
 - Always being a participant in athletic and extra-curricular activities
 - Lead college football team in major inter university championships
- **Senior Secondary - 12th From CBSE BOARD – 2012**
 - **Secondary – 10th From CBSE BOARD – 2010**

Technical skills:

Cloud –

Azure fundamentals

- Cloud Computing Concepts: Understanding the definition and benefits of cloud computing, including high availability, scalability, elasticity, agility, disaster recovery, and the shared responsibility model.
- Cloud Service Types (IaaS, PaaS, SaaS): define and differentiate between Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), and Software-as-a-Service (SaaS), and identify suitable use cases for each type.

- Core Azure Architectural Components: Understanding Azure regions, regional pairs, availability zones, datacenters, resource groups, subscriptions, and management groups, as they relate to high availability, redundancy, and disaster recovery.
- Azure Compute Services: Familiarize with Azure Virtual Machines, Virtual Machine Scale Sets, Azure App Services, Azure Container Instances (ACI), Azure Kubernetes Service (AKS), and Azure Functions (serverless computing).
- Azure Networking Services: Understanding the basics of Azure networking, including Virtual Networks (VNets), subnets, endpoints, peering, Network Security Groups (NSGs), VPN Gateway, and ExpressRoute.
- Azure Storage Services: different types of Azure storage (Blob, Disk, File, Queue, and Table storage), storage tiers, redundancy options (LRS, ZRS, GRS, RA-GRS), and data migration options.
- Azure Identity, Access, and Security: Understanding Azure Active Directory (Azure AD), authentication methods like Multi-Factor Authentication (MFA) and Single Sign-On (SSO), Azure Role-Based Access Control (RBAC), and security tools like Azure Security Center and Key Vault.
- Azure Governance and Compliance Features: familiar with Azure Policy, Azure Blueprints, resource locks, Azure Trust Center, and compliance certifications like ISO and GDPR.
- Azure Cost Management and Service Level Agreements (SLAs): Understanding factors that affect costs, utilize the pricing and TCO calculators, and be familiar with Azure Cost Management and the purpose and impact of Azure SLAs.
- Azure Management Tools: familiar with the Azure portal, Azure CLI, Azure PowerShell, Azure Resource Manager (ARM) templates, Azure Monitor, Azure Advisor, and Azure Service Health for managing and monitoring Azure resources

Azure Devops

- DevOps Strategy and Implementation: Understanding how to design and implement a comprehensive DevOps strategy, including process and communication workflows.
- Source Control Management: Mastering advanced concepts in source control with Git, including branching strategies, pull request workflows, repository configuration, and integration with Azure Pipelines. This also includes managing large files and repository scaling.
- Continuous Integration (CI): designing and implementing CI solutions using Azure Pipelines and GitHub Actions, incorporating build automation, code quality checks, and build pipeline optimization.
- Continuous Delivery (CD) and Release Management: Design and implement a release strategy, including deployment patterns (e.g., blue/green, canary), release gates, approvals, and minimizing downtime during deployments.
- Infrastructure as Code (IaC): Understanding and implementing IaC solutions using Azure Resource Manager (ARM) templates, Bicep, or Terraform to manage and deploy infrastructure consistently.
- Dependency Management: Designing and implementing strategies for managing packages and dependencies, including package management tools like Azure Artifacts and GitHub Packages.
- Security and Compliance in DevOps: Developing and implementing a security and compliance plan for DevOps pipelines, including authentication, authorization (Managed Identities, Service Principals), secrets management (Azure Key Vault), and automated scanning for vulnerabilities, dependencies, and code issues.
- Instrumentation and Monitoring: Implementing an instrumentation strategy, including logging, telemetry (Application Insights), and monitoring solutions (Azure Monitor) to gain insights into application performance and health.

- Site Reliability Engineering (SRE) Principles: Understanding SRE concepts, including designing for reliability, performance, scalability, and disaster recovery.
- Troubleshooting and Optimization: Developing strong troubleshooting skills for diagnosing and resolving issues in build and release pipelines, as well as optimizing pipeline performance, cost, and reliability

Azure Admin

- Manage Microsoft Entra ID: Learning to create and manage users, groups, licenses, and external users. Understand how to implement RBAC (Role-Based Access Control) for resource access.
- Implement Azure Policy and Governance: Mastering configuring Azure Policy to enforce standards and apply tags to resources. Understand resource locks and managing subscriptions.
- Manage Azure Storage Accounts: Learning to create and configure storage accounts, including redundancy options and access keys. Understand Shared Access Signatures (SAS) for secure access.
- Configure Azure Blob and File Storage: Knowing how to create and manage blob containers and file shares. Understand storage tiers (Hot, Cool, Archive) and lifecycle management.
- Deploy and Manage Azure Virtual Machines (VMs): proficient in creating, configuring, and managing VMs, including disk encryption, resizing, and availability options (Availability Zones, Availability Sets).
- Implement and Manage Azure Networking: Learning to create and manage Virtual Networks (VNETs), subnets, and Network Security Groups (NSGs). Understand public and private IP addresses and troubleshooting network connectivity.
- Configure Intersite Connectivity: Understanding how to configure VNet peering, VPN gateways, and ExpressRoute for connecting Azure to on-premises environments.

- Implement Load Balancing and Application Gateway: Knowing when to use Azure Load Balancer (Layer 4) versus Azure Application Gateway (Layer 7) and how to configure them for different scenarios.
- Monitor Azure Resources: Learning to use Azure Monitor for collecting metrics and logs, creating alerts, and querying logs with Log Analytics. Understand how to monitor VMs, storage, and networks using Azure Monitor Insights.
- Implement Backup and Recovery: Mastering configuring Azure Backup for VMs and other resources. Understand Azure Site Recovery (ASR) for disaster recovery and performing failovers. You also need to be familiar with creating and managing Recovery Services vaults and backup policies.

Azure Solution Architect

- Azure Well-Architected Framework Pillars: Mastering the five pillars of the Azure Well-Architected Framework – reliability, security, cost optimization, operational excellence, and performance efficiency
- Identity and Access Management Design: Understanding how to design authentication and authorization solutions using Microsoft Entra ID (formerly Azure Active Directory), Role-Based Access Control (RBAC)
- Network Architecture Design: expertise in designing robust and scalable network solutions, including Azure Virtual Networks (VNETs), subnets, Network Security Groups (NSGs), VPN gateways, ExpressRoute, firewalls, and load balancing (Azure Load Balancer, Azure Application Gateway, Azure Front Door).
- Data Storage and Integration Design: selecting and designing appropriate data storage solutions, encompassing relational (Azure SQL Database, Azure Database for MySQL), non-relational (Azure Cosmos DB, Azure Table

Storage), file (Azure Files), and large-scale analytical storage (Azure Data Lake Storage Gen2)

- **Compute Solutions Design:** Mastering the ability to choose the right compute services for various workloads, including Azure Virtual Machines (VMs), Azure App Service, container-based solutions (Azure Container Instances, Azure Kubernetes Service), and serverless computing (Azure Functions, Azure Logic Apps).
- **Business Continuity Solutions (BCS):** Designing for high availability and disaster recovery, including selecting appropriate strategies, implementing Azure Backup, Azure Site Recovery, and defining RTO (Recovery Time Objective) and RPO (Recovery Point Objective) requirements.
- **Azure Governance and Compliance:** Understanding and implementing Azure governance features, including Azure Policy, Azure Blueprints, resource locks, and management groups, to ensure compliance with organizational and regulatory requirements.
- **Monitoring and Logging Solutions:** Designing comprehensive monitoring and logging solutions using Azure Monitor, Log Analytics, Application Insights, and Azure Sentinel for proactive threat detection.
- **Application Architecture Design:** Architect application solutions that leverage Azure services for high availability, resiliency, and scalability. This includes designing for application components, caching, messaging, event-driven architecture, and API integration.
- **Migration Strategies:** Understand how to assess and interpret on-premises servers, data, and applications for migration to Azure,

DevOps

- Managing all Production/Staging server hosted on AZURE physical and virtual environment based on LINUX.

- Working on AZURE cloud creation of VPC, NAT Gateway and launching instances as per requirements.
- Working knowledge on IAM for identity on Access management for AWS using Multifactor authentication.
- Managing security groups for the instances.
- Using services such as snapshot creation, volumes, backup and restore in VMs.
- Managing docker container and images.
- Deployments of build through Jenkins and Jenkins pipelines.
- Installation of GIT on VM and pushing codes to GitHub
- Code management on GIT and repo management
- Configured continuous Integration (CI) tools Jenkins and virtual servers.
- Experienced in Tomcat and Apache web servers for deployments.
- Installation, configuration, and management of Puppet server agent setup.
- Deploy changes using manifest in puppet server. Virtual hosting of Websites on Apache server.
- Using ServiceNow as main ticketing tool for daily issues and outages

GIT/GITHUB

- concepts of the Git source control system
- Step through the entire Git workflow
- Compare the different states in Git and compare between branches and commits
- Manage files with Git (move, rename, delete) and update files managed outside Git
- Create and fork repositories on GitHub and push changes back after working after working on them locally

KUBERNETES

- Orchestrated multi-cloud or hybrid Kubernetes deployments: Describe your experience in designing, deploying, and managing Kubernetes clusters

across different cloud providers (e.g., AWS, Azure, GCP) or a combination of cloud and on-premise environments.

- Automated CI/CD pipelines for Kubernetes deployments: Highlight your skills in building and optimizing Continuous Integration/Continuous Delivery pipelines using tools like Jenkins, GitLab CI/CD, Argo CD, or Flux, resulting in faster release cycles and reduced deployment errors.
- Implemented GitOps workflows for Kubernetes management: Showcase your expertise in using GitOps principles and tools like Argo CD or Flux to manage and automate Kubernetes cluster configurations and deployments.
- Designed and implemented Kubernetes-based solutions for scaling and high availability: Mention your experience in building resilient and scalable Kubernetes architectures, potentially utilizing services meshes (e.g., Istio) and other related technologies.
- Optimized Kubernetes cluster performance and resource utilization: Detail your efforts in ensuring efficient resource allocation and cost reduction, potentially utilizing techniques like horizontal and vertical pod autoscaling, resource requests and limits, and node autoscaling.
- Implemented and enforced robust Kubernetes security policies: Describe your contributions to securing Kubernetes environments, including RBAC (Role-Based Access Control), network policies, vulnerability scanning, and secrets management.
- Troubleshoot and resolved complex issues in Kubernetes environments: Provide specific examples demonstrating your problem-solving skills in diagnosing and resolving issues related to Kubernetes infrastructure, networking, and application performance.
- Configured and maintained monitoring and logging solutions for Kubernetes: Highlight your experience with tools like Prometheus, Grafana, and the ELK stack (Elasticsearch, Logstash, Kibana) to ensure comprehensive observability of Kubernetes clusters and applications.
- Mentored junior team members and fostered collaboration: Showcase your leadership skills by mentioning instances where you provided guidance,

training, or mentorship to junior engineers on Kubernetes and cloud-native practices.

- Contributed to open-source Kubernetes projects or community initiatives: If applicable, mention your contributions to the Kubernetes ecosystem, such as developing custom controllers, submitting pull requests, or participating in special interest groups (SIGs).

DOCKER

- Using Docker, Compose and Kubernetes on your machine for better software building and testing.
- Learning of Docker and Kubernetes official tools
- Build Swarm and Kubernetes clusters for server deployments
- skills to build development environments with your code running in containers
- best practices for making Dockerfiles and Compose files

Terraform

- **Infrastructure as Code (IaC):** Proficient in writing and managing infrastructure using Terraform.
- **Cloud Provider Integration:** Experienced with AWS, Azure, Google Cloud, and other cloud platforms for resource management.
- **Resource Provisioning:** Skilled in creating and configuring VMs, networks, storage, and other resources using Terraform.
- **State Management:** Managing Terraform state files for accurate and consistent deployments.

- **Module Development:** Writing reusable and modular Terraform code for different environments.
- **Terraform Workspaces:** Using workspaces to manage multiple environments and configurations.
- **Version Control Systems:** Integrating Terraform with Git, GitHub, Bitbucket, or other version control systems.
- **CI/CD Integration:** Implementing Terraform in CI/CD pipelines for automated infrastructure deployments.
- **Collaboration Tools:** Utilizing Terraform Cloud or Enterprise for team collaboration and remote state management.
- **Security Best Practices:** Applying security best practices in Terraform configurations.
- **Monitoring and Optimization:** Monitoring Terraform-managed infrastructure and optimizing performance.

Prometheus

- **Prometheus Setup and Configuration:** Installing and configuring Prometheus for monitoring systems and services.
- **Custom Metrics Collection:** Creating and exposing custom metrics for various applications and services.
- **Querying Prometheus with PromQL:** Writing and optimizing PromQL queries to analyze and visualize data.
- **Alerting and Notification Management:** Configuring alerting rules and managing notifications using Alertmanager.
- **Integration with Visualization Tools:** Integrating Prometheus with visualization tools like Grafana for creating dashboards and visualizing metrics.
- **Scaling and High Availability:** Implementing best practices for scaling Prometheus and ensuring high availability.
- **Performance Tuning:** Optimizing Prometheus performance for large-scale environments.
- **Exporters and Integrations:** Utilizing and configuring various Prometheus exporters for different systems and services.

- **Monitoring Kubernetes Clusters:** Using Prometheus to monitor Kubernetes clusters and applications.
- **Troubleshooting and Issue Resolution:** Identifying and resolving issues related to Prometheus monitoring.

Grafana

- **Grafana Setup and Configuration:** Installing and configuring Grafana for data visualization.
- **Dashboard Creation:** Designing and building interactive, real-time dashboards for monitoring various metrics.
- **Data Source Integration:** Connecting Grafana with multiple data sources such as Prometheus, InfluxDB, Graphite, Elasticsearch, and more.
- **Custom Visualizations:** Creating custom panels and using plugins to extend Grafana's visualization capabilities.
- **Alerting and Notification Management:** Setting up alerting rules and managing notifications for critical events.
- **User Authentication and Access Control:** Configuring user roles and permissions to ensure secure access to dashboards.
- **Scripting and Automation:** Using Grafana APIs for scripting and automating dashboard creation and updates.
- **Performance Tuning:** Optimizing Grafana for large-scale environments and ensuring smooth performance.
- **Version Control:** Managing dashboards and configurations using version control systems like Git

LINUX Administration

- Linux installation and Administration.
- Server BIOS Management

- Automating Daily tasks
- User and group management
- Server Recovery working with single user mode and rescue mode, initrd, grub and kernel recovery
- Linux Performance management- Managing top, sar, iostat and free utility
- YUM Management - RPM installation and update Creation of local and FTP Yum Server
- Managing File permissions, sticky bit, SUID, SGID and ACL
- Installation and configuration of Samba server, NFS Server, Send mail.
- Installation and configuration of Apache server.
- IP Tables/Firewall Management and troubleshooting
- Process Management - Managing Process limit and Process priority
- Managing server logs as well as logs of Applications hosted on Linux server and maintaining log rotation.
- Job scheduling (AT and CRONTAB utility)
- Nagios as Monitoring tool
- Patching servers under latest OS version to make them compliant as per client requirements.
- Handling tickets outages with no or minimal downtime.
- Solving tickets as per SLA.
- Day to day application support on production Environment and technical documentation for critical prod issues.

Trouble Shooting (ITIL)

- Aware about ITIL procedure and have real time experience on working these steps for timely solution to any problem encountered.
- Proactive Problem Identification
- Problem Categorization and Prioritization
- Problem Diagnosis and Resolution
- Problem and Error Control
- Problem Closure and Evaluation
- Major Problem Review

- Problem Management Reporting and getting it documented for further reference and solution

SQL

- Complete literal SELECT statements.
- Complete basic SELECT statements querying against a single table
- Filter data using the WHERE clause
- Sort data using the ORDER BY clause
- Learn about database design and normalization
- Using joins, return columns from multiple tables in the same query
- Understand the difference between the INNER JOIN, LEFT/RIGHT OUTER JOIN, and FULL OUTER JOIN
- Complete SQL statements that use aggregate functions
- Group data together using the GROUP BY clause
- Filter groups of data using the HAVING clause

Remedy Tool

- Worked on BMC remedy tools as primary tool for Change management and incident reporting
- Knowledge of ITSM daily requirements
- Maintained and support incident reported from interfacing Applications.
- Knowledge of SLA and timely resolutions of task and action items assigned

ACHIEVEMENTS

- 3 Annual Tech Mahindra BRAVO Award recipient
- 4 successful U2L (Unix to Linux) Migration
- Pro-actively prevented 10+ outages which may cause huge business loss to the client.
- Always being a team player and get appreciated by client many times for quick and proactively resolving issues

